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1. To prove that output is less than or equal to everything in data1 and data2 after every iteration of that for loop in MysterySort, we can use induction.

Base Case: After the first iteration ($i=1$), $\text{output}[i]$ will be the smallest element between $\text{data1}[\text{left}]$ and $\text{data2}[\text{right}]$, Since both data1 and data2 are sorted and the size of the array is one. Therefore $\text{output}[1]$ will be less than or equal to all elements in data1 and data2.

Inductive Step: Assuming that after the k -th iteration, $\text{output}[i]$ is less than or equal to everything in data1 and data2. We want to show that after the $k+1$ th iteration the statement holds.

During the $(k+1)$ th iteration, we compare $\text{data1}[\text{left}]$ and $\text{data2}[\text{right}]$, Since both data1 and data2 are sorted, the smaller of the two will be in $\text{output}[i]$ and will be less than equal to everything in data1 and data2. Therefore, by induction, we can conclude that after every iteration of the for loop, $\text{output}[i]$ is ~~the~~ less than and equal to everything in data1 and data2.

2. After the second iteration output[1] will contain the smallest element between data1[~~left~~] and data2[~~right~~] which is guaranteed to be less than or equal to all elements in data1 and data2 due to ~~the~~ sorting. Similarly for output[2]. Therefore, output[1] $\leq$ output[2].

3. Using the formula for the sum of squares, we have:

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} \rightarrow ni^2 = 1^2 + 2^2 + 3^2 + \dots + n^2$$

For all $i \geq 1$, ~~we~~ we have $i^2 \leq n^2$. Thus

$$\sum_{i=1}^n i^2 = ni^2 \leq \sum_{i=1}^n i^2 = 1 \rightarrow nn^2$$

$$\sum_{i=1}^n = (n)(n^2) = n^3 \rightarrow O(n^3)$$